

Public Perceptions of AI and Art: Mapping a Cultural Shockwave

1 Introduction



Generative AI didn't just quietly enter the tech world; it kicked the door down. Almost overnight, tools like Midjourney, DALL-E, and ChatGPT went from niche research projects to everyday tools capable of creating stunning visuals and intricate stories. But as AI democratized creation, it also ignited a fiery cultural debate. Whilst that debate is philosophically, legally and morally interesting, there are also questions to be asked about the reaction to the advent of this revolutionary technology. How did we (artists, technologists, and the general public) actually react when machines started making art? Were we inspired, terrified, or just trying to sell NFTs?

In this post, we map the massive cultural vibe shift that took place during the peak of the generative AI boom. To do this, we focussed on a major source of discourse at that time - Twitter. Our motivation was the following: the sudden, overwhelming accessibility of 'creative' AI immediately polarized the internet, and we wanted to understand how and why. Our core research question asks: How did the public react to the intersection of AI and Art? and how did those responses cluster into distinct online communities?

We aren't just guessing. We have the receipts - millions of them.

The Cambrian Explosion of Generative AI

To understand the data, we first have to understand the sheer velocity of the era. The 12-month period from June 2022 to June 2023 represents a true Cambrian explosion of generative AI.

This wasn't a steady evolution; it was a rapid-fire sequence of blockbusters. We saw the broadening of DALL-E 2, the open-source shock of Stable Diffusion, early versions of Midjourney, and the absolute paradigm shift of ChatGPT's launch on November 30th. Every few weeks, a new foundational model (LLaMA, GPT-4, PaLM 2) or a massive commercial integration (New Bing, Adobe Firefly) dropped.

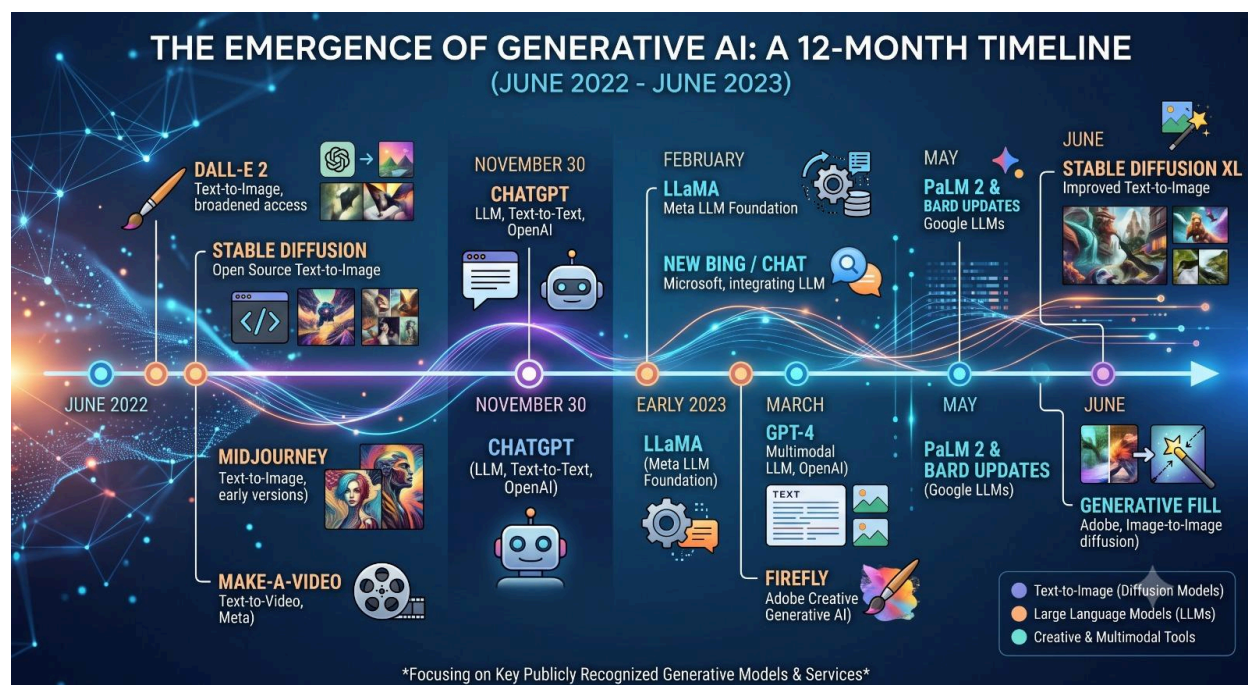


Figure: THE EMERGENCE OF GENERATIVE AI: A 12-MONTH TIMELINE (JUNE 2022 - JUNE 2023). Focusing on Key Publicly Recognized Generative Models & Services.

This timeline isn't just a list of release dates; it's the heartbeat of our dataset. While the graphic above shows a full year of development, the data we analyzed focuses on the intense 120-day window from October 31, 2022, to February 27, 2023 - the exact moment the ChatGPT shockwave hit the public consciousness.

Finding the Human Signal in 36 Million Tweets

You can't map a cultural shift with a handful of opinions, so we went big. We tapped into the Twitter API to pull every conversation happening around AI during this explosive 120-day window. We started with a massive raw pile of **36.6 million records**.

But the internet is messy. It is full of duplicates, bots, and spam. To find the true human signal, we put the data through a rigorous, multi-stage cleaning pipeline:

1. **Deduplication:** We stripped out nearly 8.5 million duplicate records.

2. **Spam & Bot Filtering:** We rigorously filtered out the noise. For example, we removed thousands of crypto bots trying to hijack AI hashtags with "#airdrop" spam.
3. **Language Detection:** We applied strict language detection to ensure we were looking at a cohesive English-language corpus, dropping French hits for "j'ai" or Spanish false positives.
4. **Text Normalization:** We stripped out URLs, @mentions, and the "RT" markers, leaving only the raw, emotional text of the user.

The result? A pristine **"AI Corpus" of 17.4 million tweets** from 4.7 million unique authors.

From there, we dug even deeper. We filtered this massive corpus against 60 specific creative keywords (like *aesthetic*, *illustration*, *copyright*, *emotion*, *plagiarism*) to extract our core **"AI + Art Corpus"**. This left us with a highly focused dataset of **3.58 million tweets** from 1.4 million authors - it's the largest curated archive of the exact moment humanity collided with generative art.

Sentiment and Emotion Profile

The clearest finding was that AI-and-art discussion was substantially more negative than general AI discussion. In the full 198,326-author analysis, mean negative sentiment was 0.330 for AI-and-art discussion and 0.159 for general AI discussion. Positive sentiment showed the opposite pattern, averaging 0.194 and 0.341 respectively. Neutral sentiment was similar, at 0.476 and 0.500.

We measured these differences using two models from [CardiffNLP](#), Cardiff University's natural language processing research group. Both models were trained to classify language in Twitter posts. The sentiment model assigns probabilities to three categories: positive, neutral and negative. Each probability ranges from 0 to 1, and the three sum to 1. Higher values indicate greater model confidence in that classification.

The emotion model scores eleven categories separately: anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise and trust. These scores also range from 0 to 1. Several emotions can receive high scores in the same post. The analysis covers post text; attached images, videos and linked content were excluded.

AI-and-art posts had higher mean scores for anger, 0.292 compared with 0.167, and disgust, 0.298 compared with 0.174. They had lower scores for joy, 0.373 compared with 0.550, and optimism, 0.288 compared with 0.402. Anticipation was similar in the two corpora, at 0.484 and 0.496. The difference in sentiment therefore accompanied a more specific pattern of higher anger and disgust and lower joy and optimism.

We also examined how sentiment and emotion changed over time. We averaged each author's scores within each week, then averaged across active authors. This gave each author equal influence on the weekly result, regardless of posting frequency.

In the week containing ChatGPT's launch, 28 November to 4 December 2022, mean positive and negative sentiment were almost equal, both rounding to 0.285. By the week beginning 12 December, negative sentiment had risen to 0.292 and positive sentiment had fallen to 0.271. That week also had the highest mean anger and disgust scores, at 0.298 and 0.304, and the lowest mean joy score, at 0.393. By the final complete week, beginning 20 February 2023, positive sentiment had risen to 0.340 and negative sentiment had fallen to 0.214.

These trends reflect both changes in what people posted and changes in who participated. The earliest weeks also contain fewer active authors, making comparisons across the period sensitive to the changing composition of the conversation.

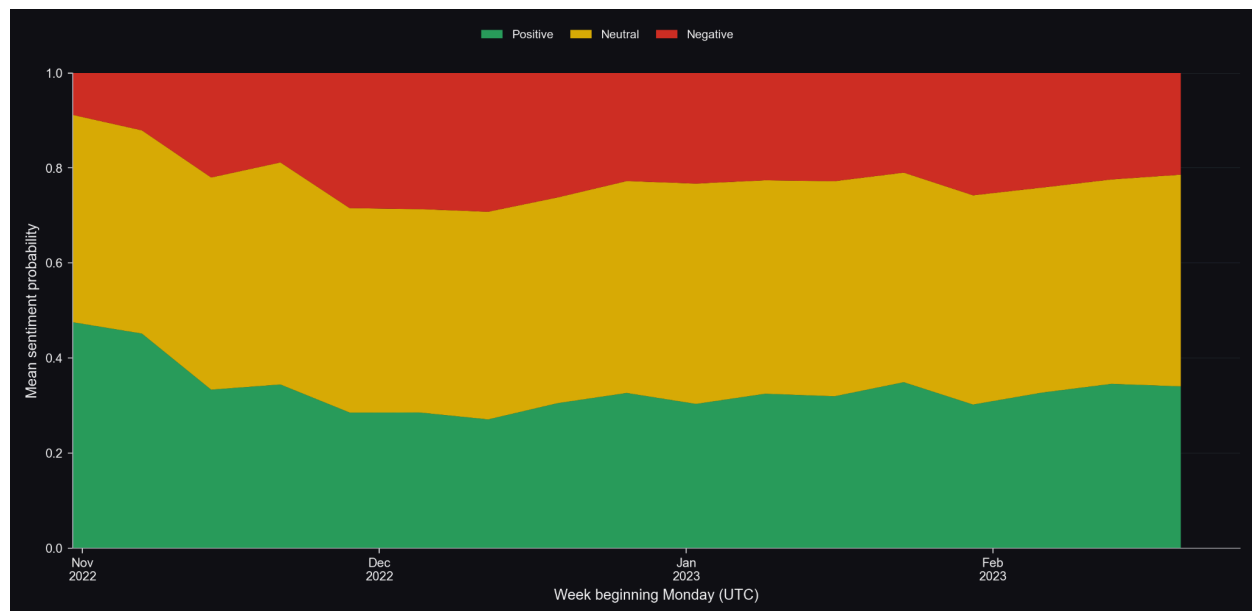


Figure: Weekly mean positive, neutral and negative sentiment probabilities, averaged within author-week before aggregation across active authors. The incomplete final week is omitted.

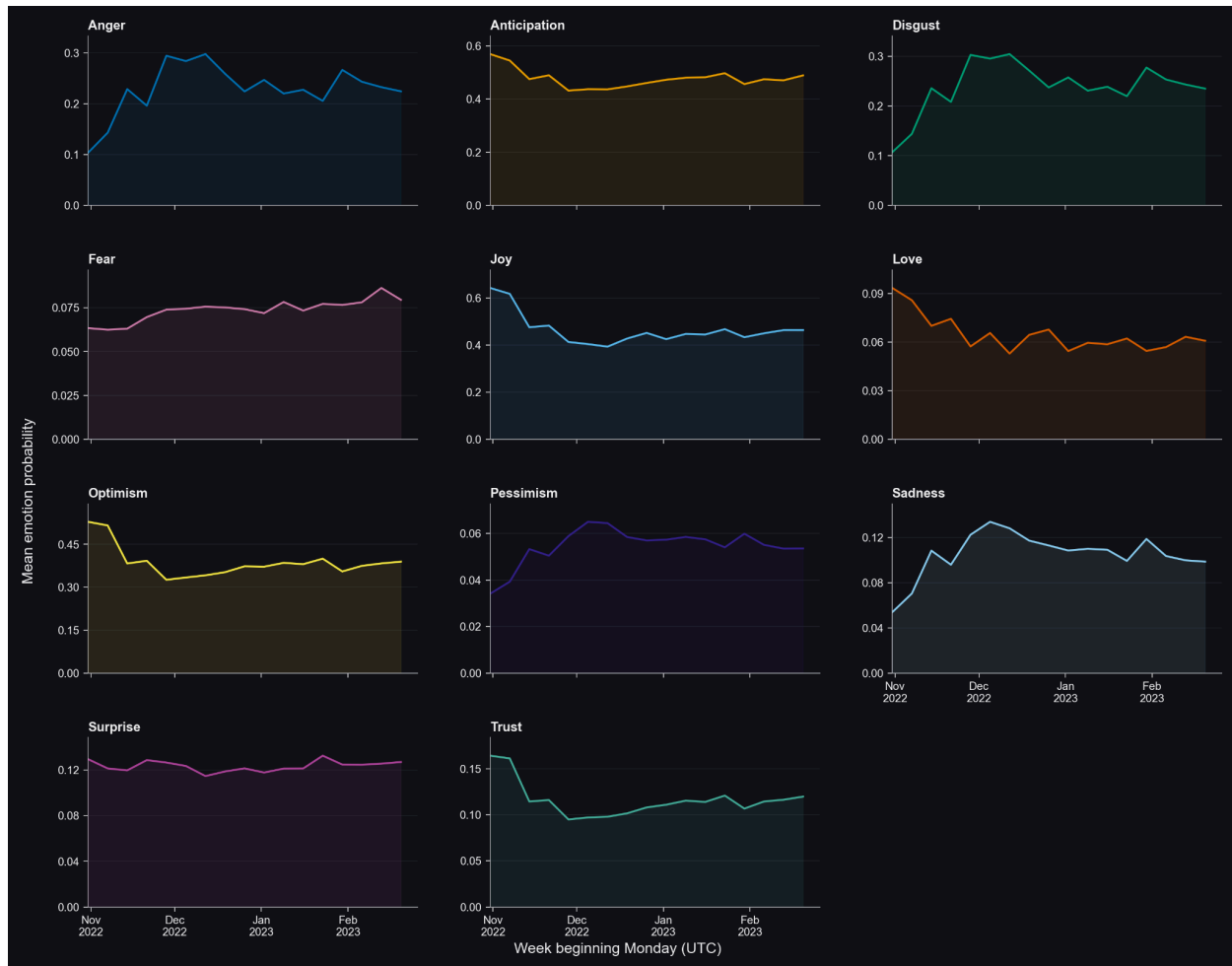


Figure: Weekly mean scores for the eleven emotion categories, using the same author-week aggregation. Each panel has its own vertical scale.

Topics

[Latent Dirichlet allocation](#), or LDA, identifies topics from patterns of words occurring together across documents. It represents each document as a mixture of topics, with each topic characterised by its own distribution of words. We combined each author's posts into one document and fitted a twelve-topic model, then named the topics by examining their characteristic terms and representative documents.

The topics covered AI Art Discourse, General AI, AI Tools/Code, Visual AI Art, Bard/LLMs, NFT/Crypto, Web3/DeFi, Marketing/Social, Tech/Programming, News/Updates, Trading/Invest and Bot/Spam. Each author received a share of membership in all twelve topics, allowing their profile to include several subjects.

Among the 198,326 matched authors, AI Art Discourse was the highest-weighted topic for 120,665, or 60.8%. General AI was the highest-weighted topic for another 38,980 authors, or 19.7%. These were the two largest concentrations of discussion in the analytical sample.

The UMAP figure projects these twelve-dimensional topic profiles into two dimensions. Each point represents an author, coloured by their highest-weighted topic. Nearby points generally have similar topic mixtures. The coloured layer is capped within each topic to keep smaller topics visible.

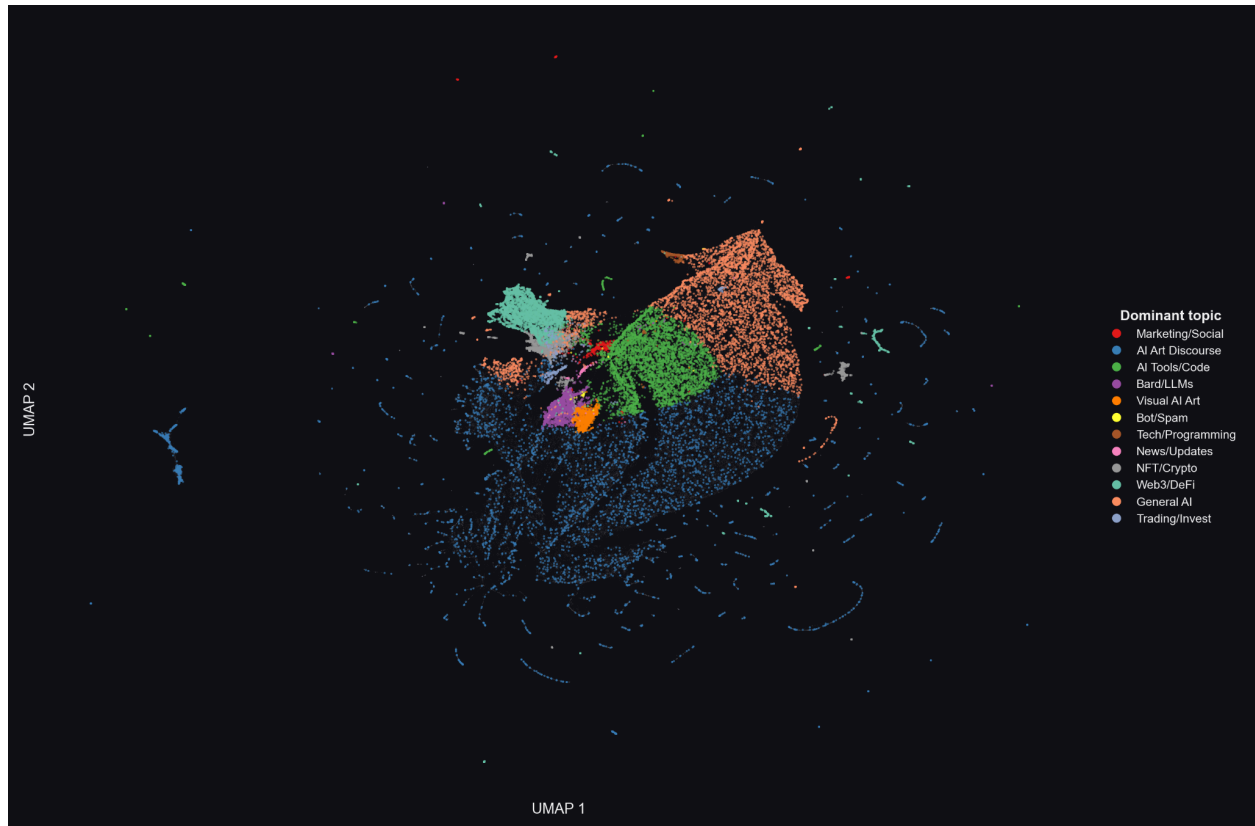


Figure: UMAP projection of author topic profiles, coloured by dominant topic. The coloured overlay is capped within each topic for visibility.

To track topics over time, we applied the same fitted model to each author's posts in each week. AI Art Discourse grew through November and early December, reaching 57.3% of the mean topic mixture in the week beginning 12 December. Its share subsequently declined, while remaining the largest topic. Across the period, it accounted for 46.6% of the author-week-weighted topic mixture. General AI and AI Tools/Code were substantial secondary topics.

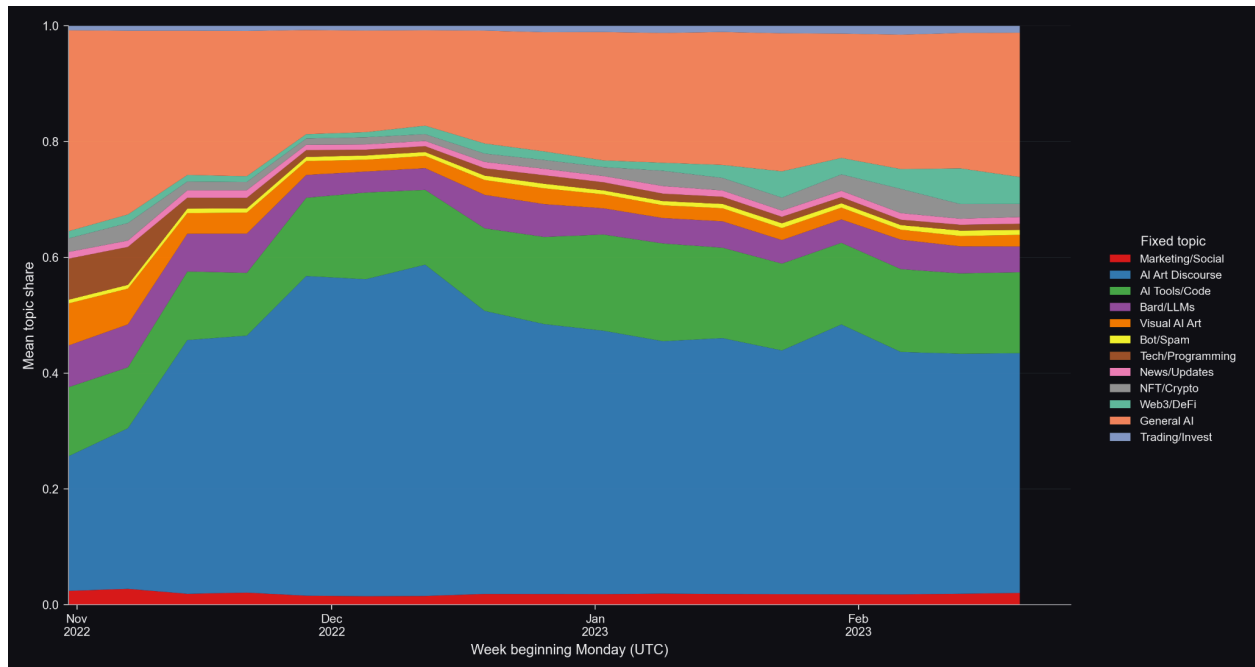


Figure: Weekly mean topic composition across active authors, calculated using the fixed twelve-topic model.

Sentiment also differed between topics. We calculated net sentiment as positive probability minus negative probability, giving a scale from -1 to $+1$. Web3/DeFi had the highest mean net sentiment, at $+0.719$, followed by NFT/Crypto at $+0.573$. AI Art Discourse was the only topic with negative mean net sentiment, at -0.143 .



Figure: Mean positive, neutral and negative sentiment probabilities by topic, alongside net sentiment.

Communities

The interaction network provides a second way to organise the discussion. Its connections record retweets, replies and quotations between authors. We identified communities using the directed Leiden algorithm, which groups authors according to these patterns of interaction. The full network contained 1,984,599 authors and 4,472,376 connections; the combined analysis used the 198,326 authors with matching topic, sentiment and emotion data.

These matched authors belonged to 946 communities. The 21 largest contained 188,765 authors, or 95.2% of the sample. General ChatGPT/AI Discussion was the largest, with 44,718 authors, followed by Anti-AI-Art Discourse with 41,574 and General ChatGPT/AI Debate with 17,422. The AI-Art & NFT Posting community contained 12,679 authors.

We labelled the largest communities using class-based TF-IDF, which highlights vocabulary distinctive to each group. This analysis covered 1.19 million deduplicated posts. We checked potentially over-specific labels against samples containing one post from each of 25 distinct authors. The resulting labels describe prevalent forms of posting within each community.

Several communities centred on practical applications, including ChatGPT for education, SEO and content marketing, data-science news, and medical imaging. Others focused on AI-art criticism, authenticity disputes, generated images and NFTs. A further set revolved around

cryptocurrency tokens, launchpads and promotional activity. The network therefore separates several forms of engagement that appear together in the broader AI conversation.

Sentiment, Topic and Community

Community membership was associated with differences in emotional tone as well as subject matter. Across the 21 largest communities, it was associated with an average of 21.1% of the variation in topic weights, 32.6% of the variation in sentiment probabilities and 27.6% of the variation in emotion scores. These estimates use bias-corrected omega-squared, a measure of how much variation is associated with differences between groups. On these measures, the association with sentiment was larger than the average association with topic.



Figure: Mean positive, neutral and negative sentiment probabilities across authors in each community.

The Anti-AI-Art Discourse and AI-Art & NFT Posting communities illustrate this relationship. AI Art Discourse accounted for 87.3% of the former’s topic mixture and 51.9% of the latter’s. Both

were heavily engaged with AI art, but their emotional profiles differed: mean negative sentiment was 0.500 in Anti-AI-Art Discourse and 0.133 in AI-Art & NFT Posting. Mean joy was 0.277 and 0.614 respectively, while disgust was 0.497 and 0.149.

For example, some posts from the Anti-AI-Art Discourse community expressed outright rejection:

“ai art is fucking garbage idgasf”

“why i block ppl: - annoying - bot - rude - support ai art (uses novelai and etc)”

Posts from AI-Art & NFT Posting included appreciation of generated images and discussion of making and selling work, along with a distinct affinity for hashtags.

“ai art.... love it or hate it... you just have to admit some of it is just amazing!!! 🤖”

“gm ☕️🤖 have a great day ☀️ my first ai works, based on my photographs. what do you think, mint or not? 👁️ #nftcommunity #nftphotography #nftfam #nfts #filmphotography”

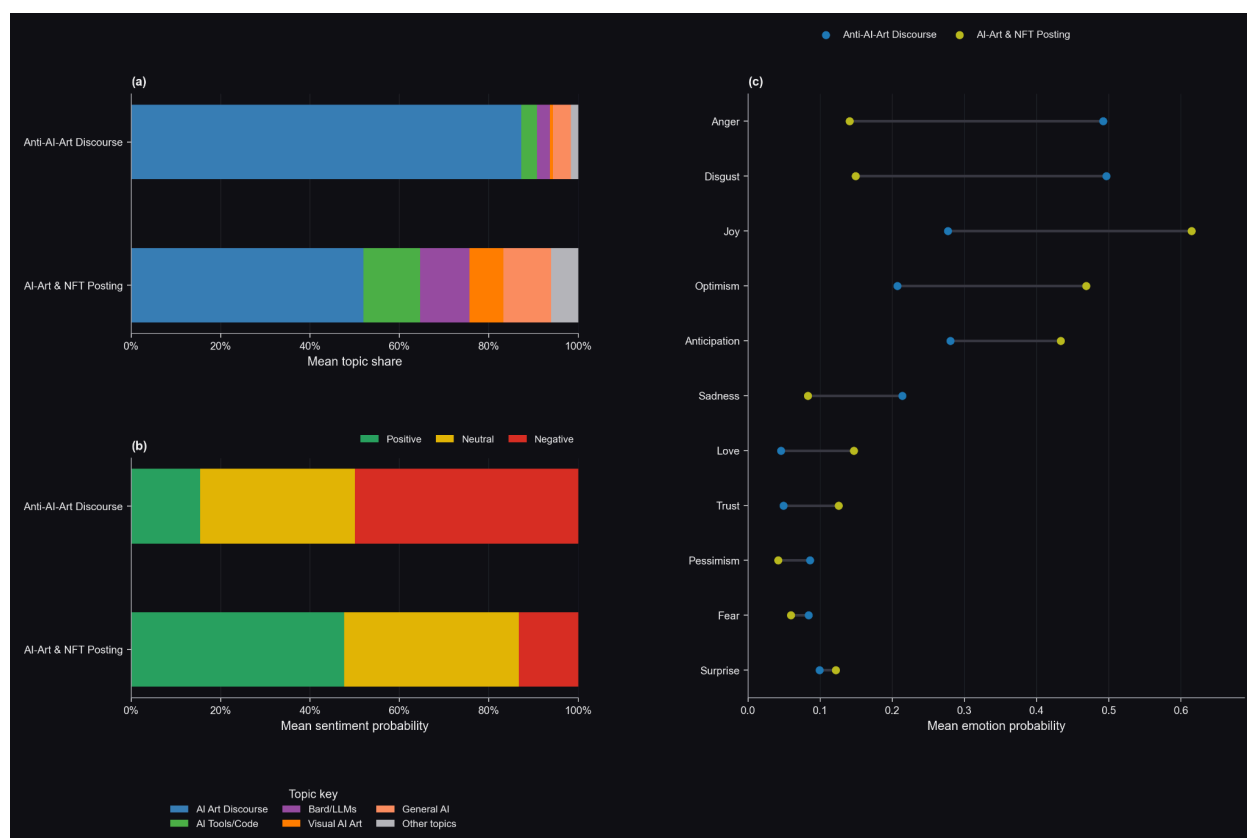


Figure: Comparison of Anti-AI-Art Discourse and AI-Art & NFT Posting across topic composition, sentiment and emotion.

Community membership offers another lens for analysing these differences. Some of the highest positivity appeared in cryptocurrency and token-promotion communities. AI/Crypto Launchpads had a mean positive sentiment probability of 0.769, accompanied by high trust, joy and optimism scores. Web3/DeFi accounted for a share of its topic mixture roughly fourteen

times the average across matched authors. Other positive communities included AI-token and launchpad posting and AI-blockchain/token promotion.

Their posts frequently used repeated formulations such as “best project”, “very good project” and “bullish on this project”. High positive sentiment here accompanied a vocabulary of endorsement, investment and anticipated success. This gives the positive side of the corpus several distinct components: enthusiasm for creative work, interest in AI applications, and promotion of tokens and commercial projects.

The topic-enrichment heatmap shows which subjects were disproportionately common in each community. Its $\log_2$ scale expresses relative prevalence: +1 means twice the overall topic share, while -1 means half. AI/Crypto Launchpads had a Web3/DeFi enrichment of +3.82, corresponding to the roughly fourteenfold concentration noted above.

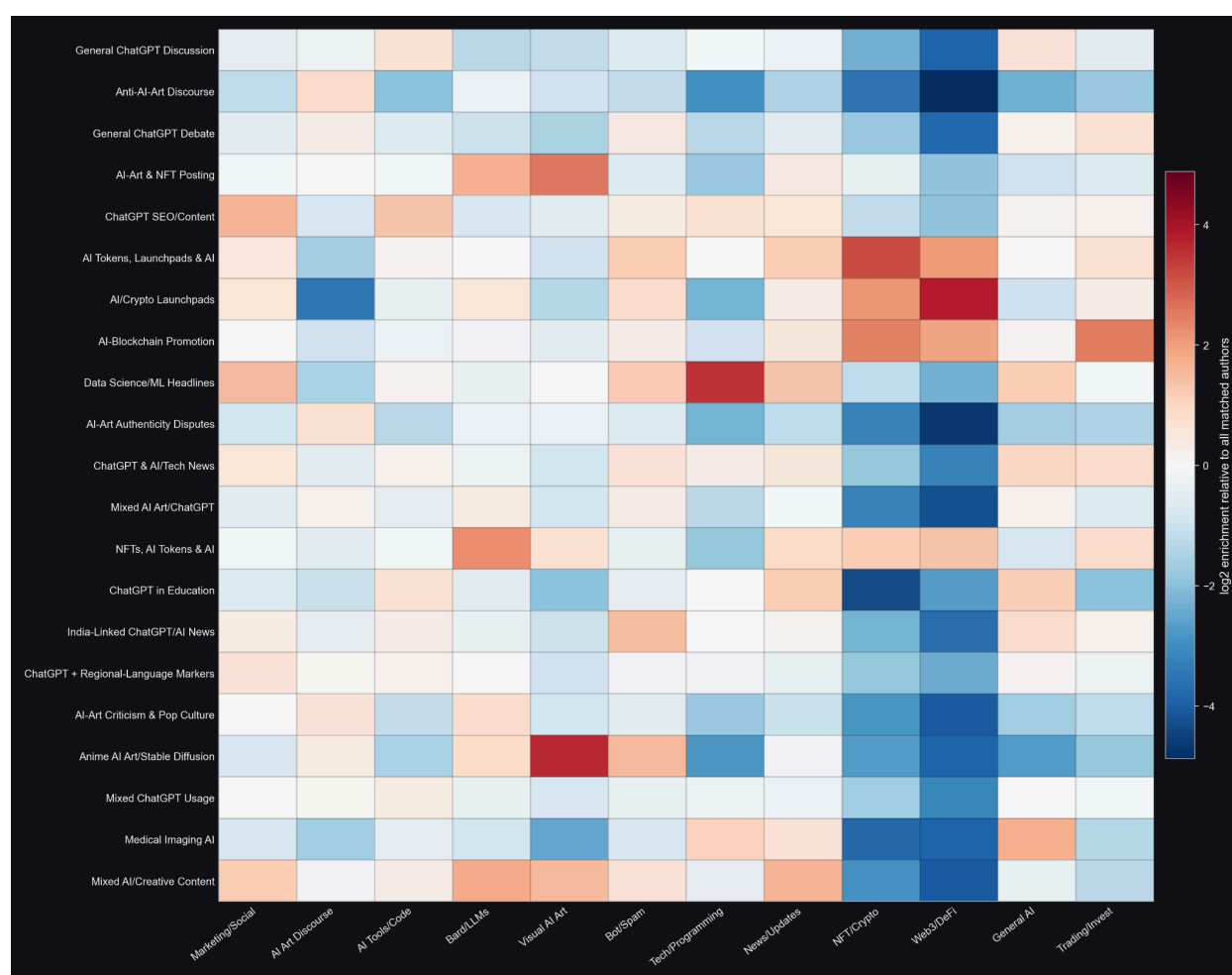


Figure: Topic enrichment by community, expressed as $\log_2$ ratios relative to all matched authors.

We also calculated topic-weighted net sentiment within each community, weighting authors' overall sentiment scores by their membership in each topic. The most negative supported combination was Anti-AI-Art Discourse $\times$ AI Art Discussion, at -0.368; the most positive was AI/Crypto Launchpads $\times$ Web3/DeFi, at +0.798. Each cell summarises the overall sentiment of

authors associated with that community and topic. Measuring sentiment specifically within posts about each topic would require a separate post-level analysis.

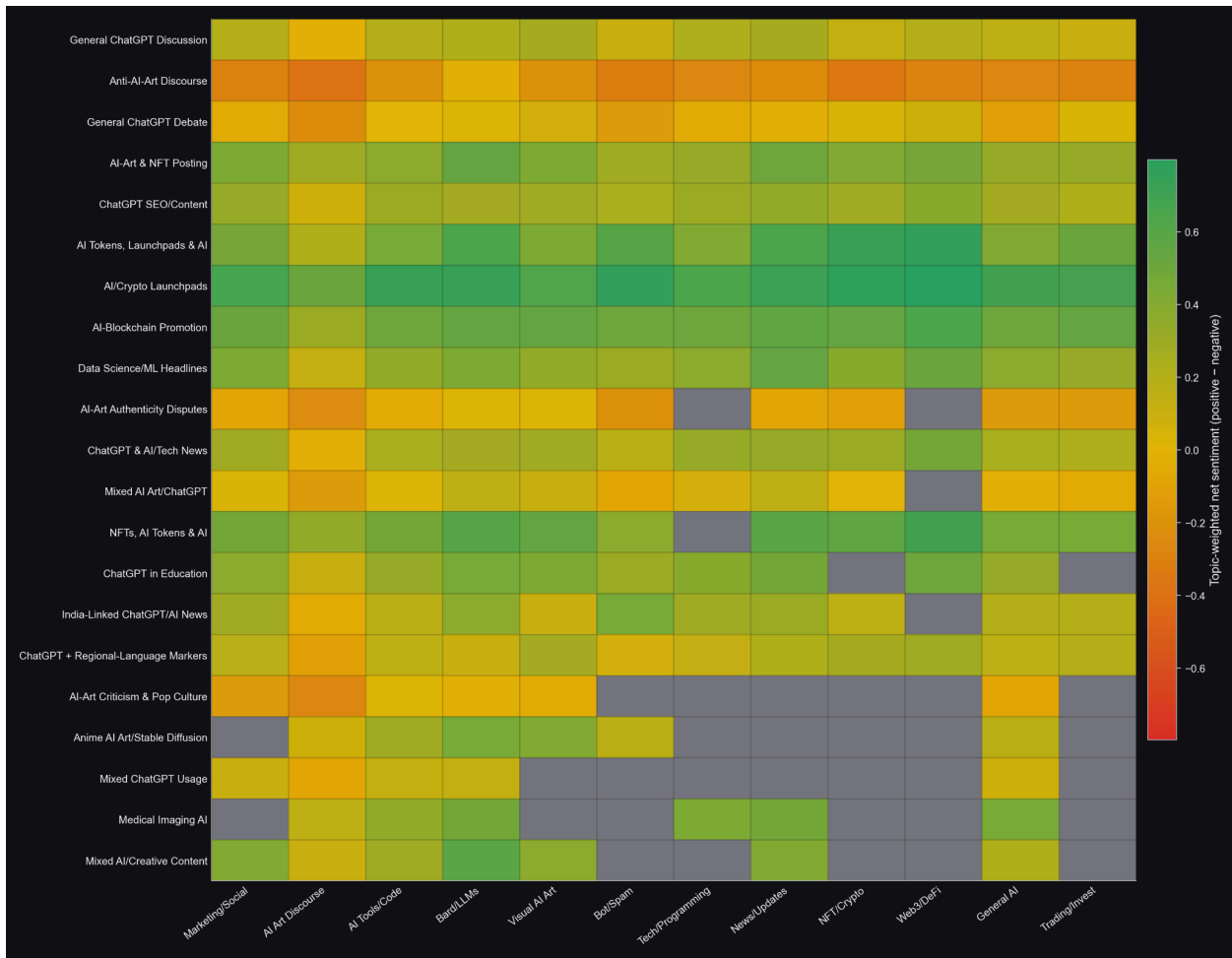


Figure: Topic-weighted net sentiment by community. Grey cells have insufficient support and are masked.

Emotion scores also varied among authors within each community. The final heatmap measures this spread using standard deviations; the median across the 231 community-emotion combinations was 0.103.

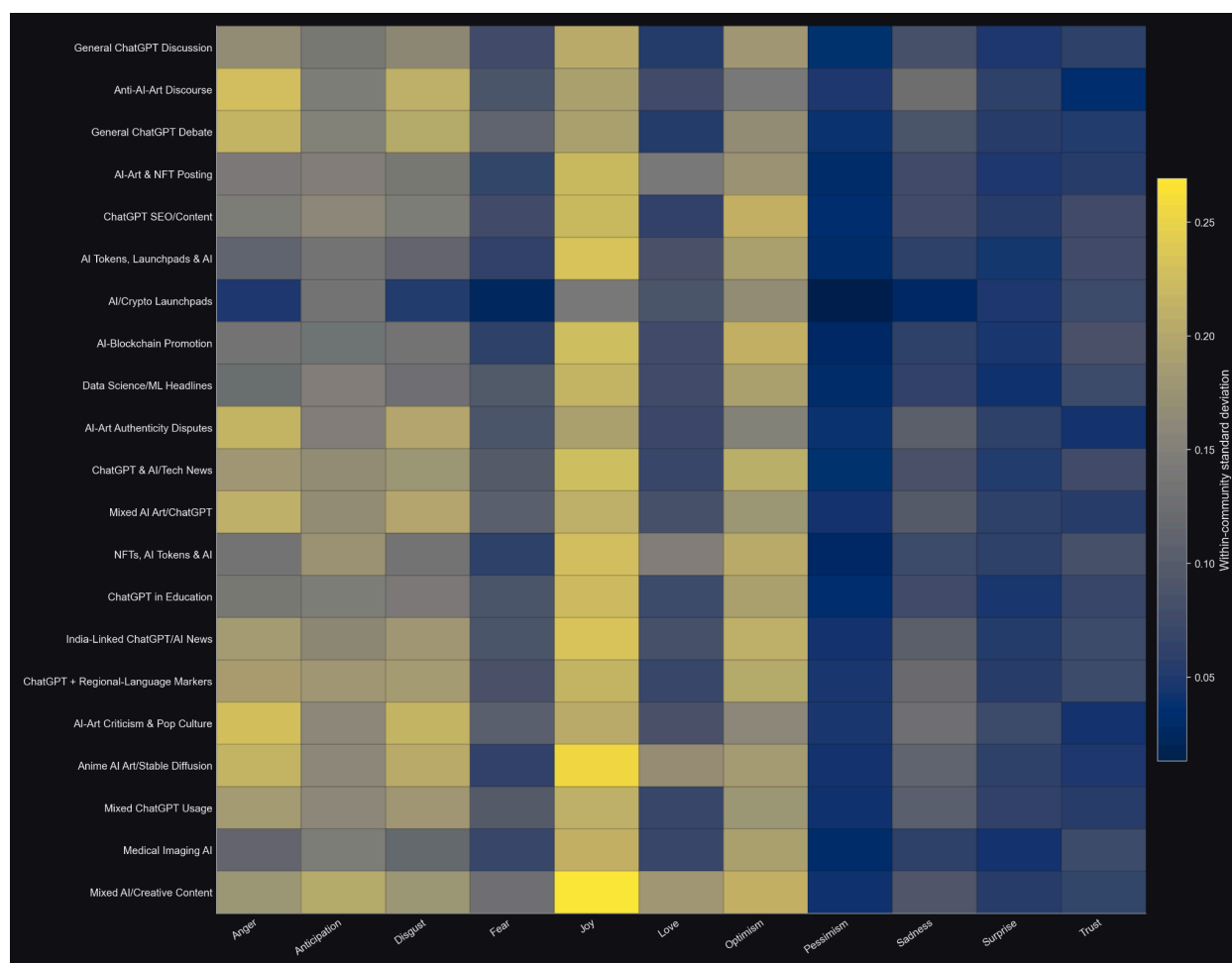


Figure: Within-community standard deviation of each emotion score.

Exploring the Network

The [interactive author map](#) displays all 198,326 matched authors. Their positions were calculated from 853,133 connections using the Distributed Recursive Layout algorithm. The layout uses interaction data, while colour displays community membership, topics, sentiment or emotion. Switching between colour schemes shows how these attributes are distributed across the same network.

Clicking an author reveals their community, topic profile, sentiment and emotion scores, and an example post selected for informative content. An optional three-dimensional view displays the 187,095 authors in the largest connected component. Readers can use the map to examine the concentrations of AI-art criticism, image-sharing and crypto promotion, then inspect individual posts within them.



Figure: Screenshot of the [interactive author map](#). Author positions reflect their connections in the interaction network; colours can display communities, topics, sentiment or emotions.

The Larger Picture

AI-art discussion was more negative than general AI discussion, with higher anger and disgust and lower joy and optimism. Within that broad pattern, the network separated several forms of participation: opposition to generated art, enthusiastic image-sharing, and involvement in NFT markets. The contrast between the two AI-art communities shows how a common subject can attract different forms of engagement, with different emotional profiles.

The positive communities were also varied. Some centred on creative work, education or technical applications. Others used the highly affirmative language of cryptocurrency and token promotion. A high positive sentiment score can therefore accompany appreciation of an image,

interest in a useful tool, or an endorsement intended to attract collectors and investors. The content and social context help distinguish these meanings.

Community membership was associated more strongly with sentiment than with topic weights on average. This suggests that emotional tone was an important feature of how the conversation was organised: different groups discussed overlapping subjects in different ways. The weekly results add a temporal dimension, with negative sentiment, anger and disgust peaking in mid-December before the conversation becomes more positive towards February.

Conclusion?

Potential bubbles?

Next steps - what concepts people are invoking (not just what topics)?



Sen. Bernie Sanders

@SenSanders · [Follow](#)



Pope Leo is right.

Art is not some kind of superfluous luxury that can be eliminated from our lives. It is a crucial part of the human experience.

We cannot let Big Tech oligarchs destroy it.



Pope Leo XIV @Pontifex

In this era of artificial intelligence, it is becoming urgent to distinguish human art from what machines produce. There is an ontological difference, even before an aesthetic one, between art and what a machine can generate through statistical calculation based on millions of

12:39 AM · Oct 5, 2026



6.9K



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